Concern Resolution Process

QC Story

QC STORY PROJECT

PROJECT TITLE :-

To Reduce the Defect of Jack mounting bracket bolt not fit in Scorpio vehicles by 80%.

Mahindra and Mahindra Ltd , NASHIK Plant, Scorpio TCF.

	Plan	Actual
Start Date	**01.11.18**	**01.11.18**
Completion Date	**02.08.19**	**20.12.19**

Current Status: Completed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

Cross Functional Team

No	Description	Name	Dept	Phone
1	**Team Leader**	Rahul Gadekar	PRD	9822855737
2	**Mentor**	Rahul P. Narkar	PU HEAD	9881080299
3	**Member 1**	Amit Tile	PRD	9823147429
4	**Member 2**	Patil Swapnil	PVT	9284534665
5	**Member 3**	Santosh Jadhav	MQA	9423309400
6	**Member 4**	Hemant Khare	PRD	9766874268

1. Select the Theme

Brief description of Problem

What : Jack mounting bracket bolt not fitted.

Who: jack mounting bracket fitment associate .

When: From Aug-2018.

Where: Jack fitment stage.

Why : Chronic issue leading to FRC loss & major Rework.

How: Avg 53 concerns per month

Scorpio Jack mounting bolt not fitted

2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-

– DPR Improvement

- **FRC / FBO Improvement**
- **Offline Rework Reduction**

2. Justify the choice

Importance of Selection of Problem -

A) Jack mounting bracket bolt not fi causes rattling noise while driving

B) Part rejection cost is =
Total Rework Cost / Yr =21123.61

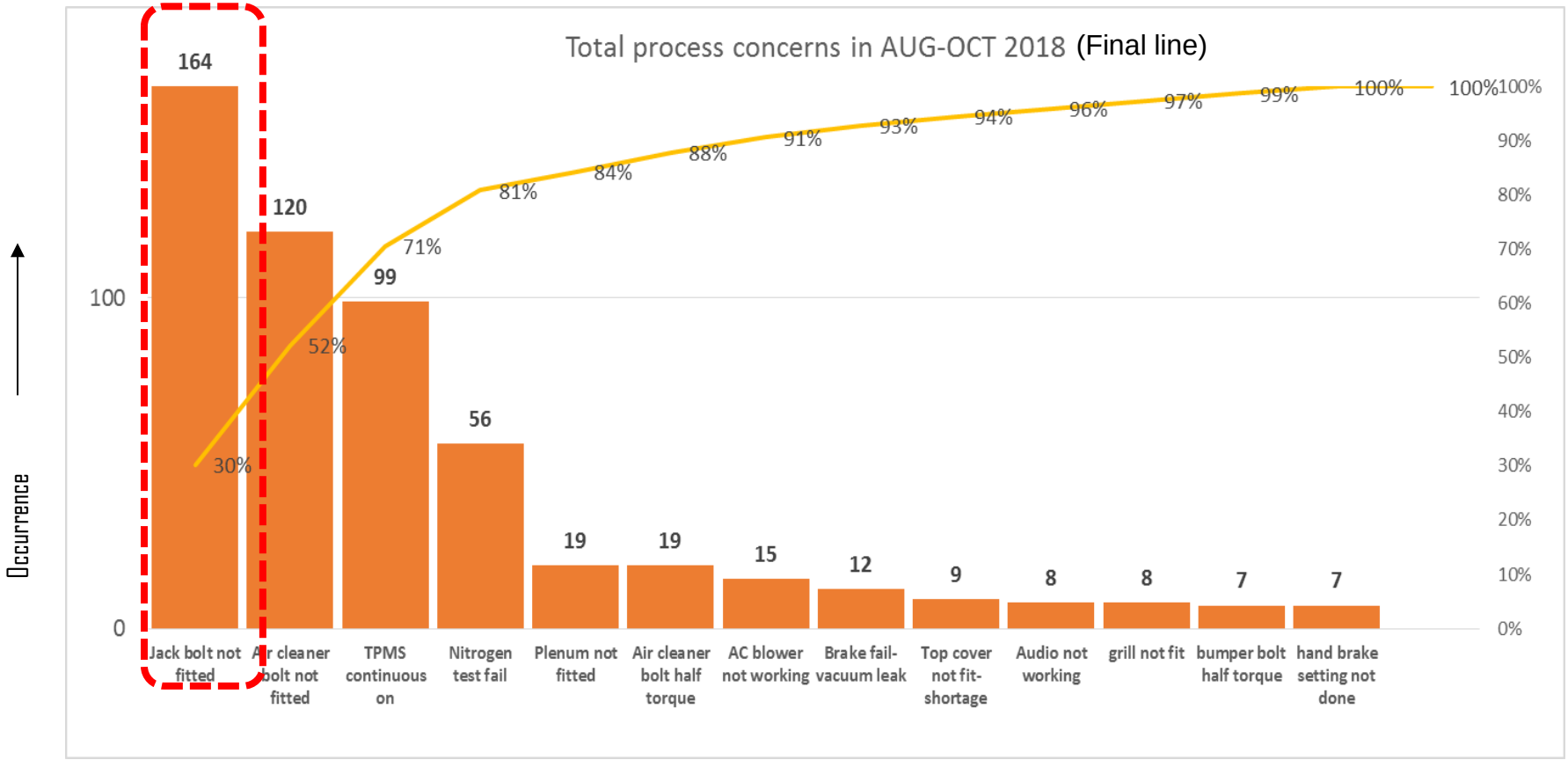
C) Average 20 vehicles reported for this concern per month

D) To improve FRC of line and ultimately shop FRC.

Cost of Poor Quality

Cost estimation for Jack mth bkt bolt change		
Description	Units	
Rework time (Approx)	Min	90
Avg Concerns/Day	Nos	1
Total Required Rework time per day	Min	90
Effective working time	Min	465
Operator required /day	nos	0.193548387
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	62.90322581
Avg no. of working days	days	25
Operator cost /month	Rs	1572.580645
Operator Cost/year	**Rs**	**18870.96774**
Bolt cost per veh	Bolts / veh.	1
Bolt cost	Rs/bolt	3.61
Daily bolts rejection	No bolts /day	2
Cost impact per bolt on change	Rs/day	7.22
Project annual volume	Rs / Year	2252.64
Total Rework cost	Rs / Year	21123.61

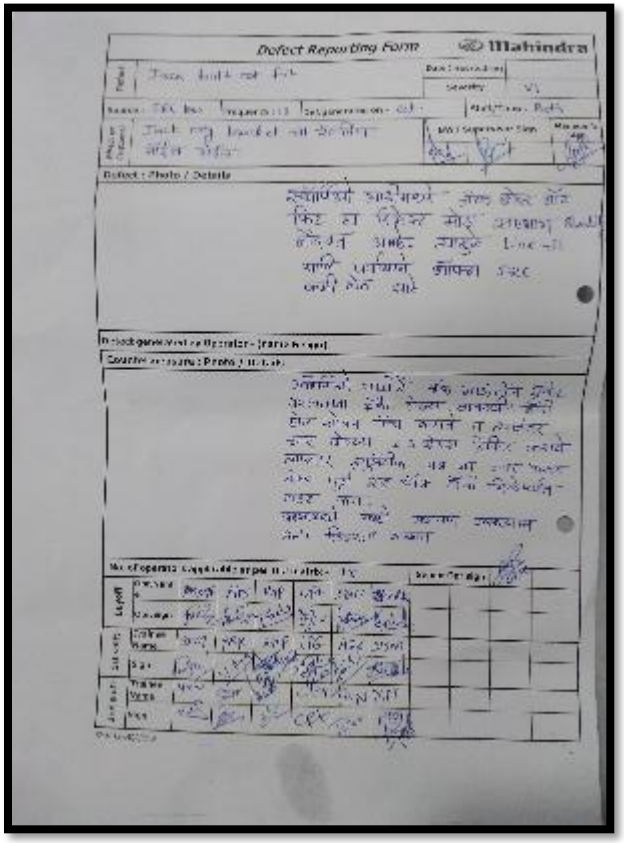
2. Justify the choice



Inference: From Pareto analysis of Scorpio it is found that Jack bolt not fitted is top PROCESS concern hence it is selected for project.

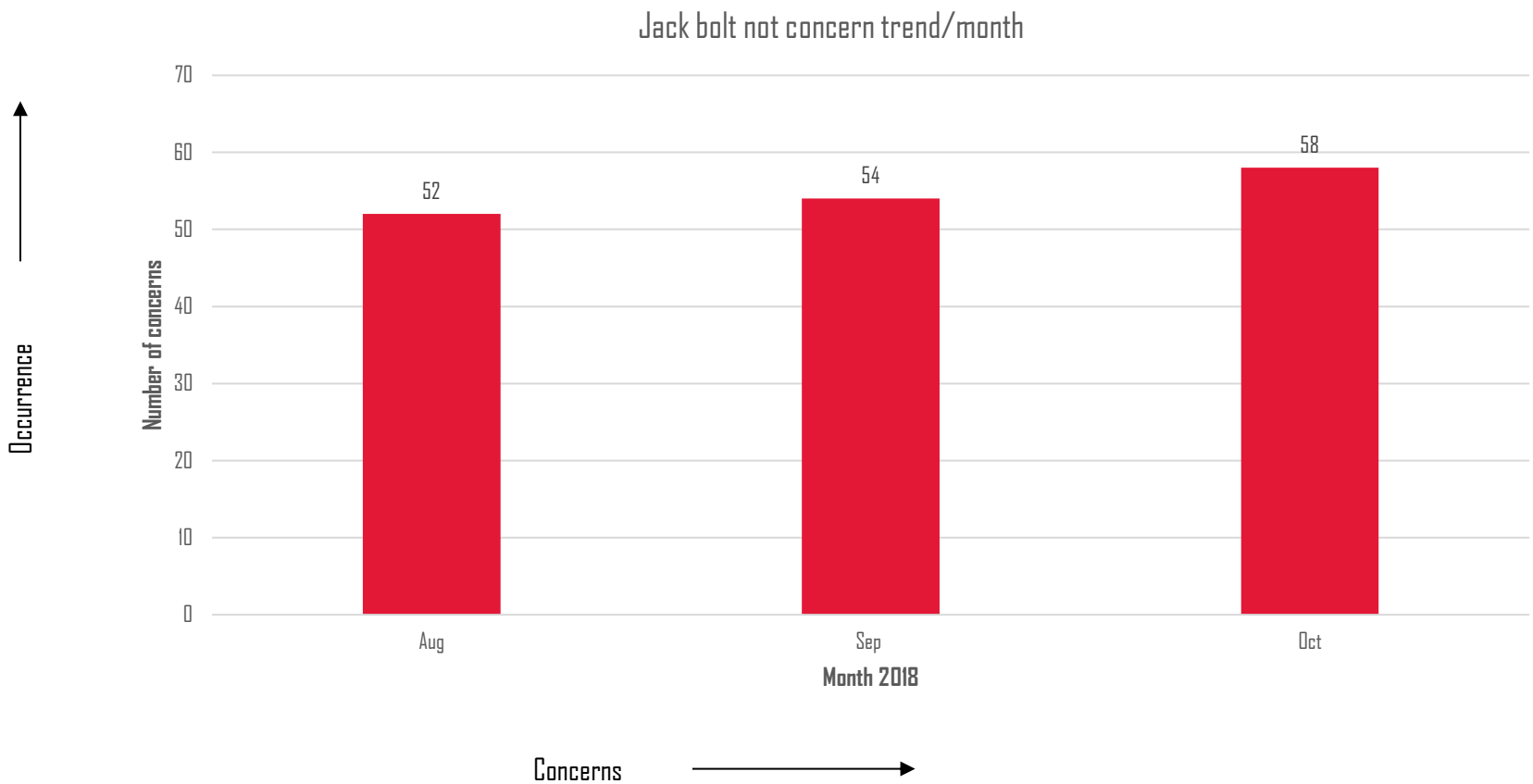
2.a Immediate Containment Action

- DRF given to stage associates for Jack bolt not fit concern communication.



3. Understand the current situation

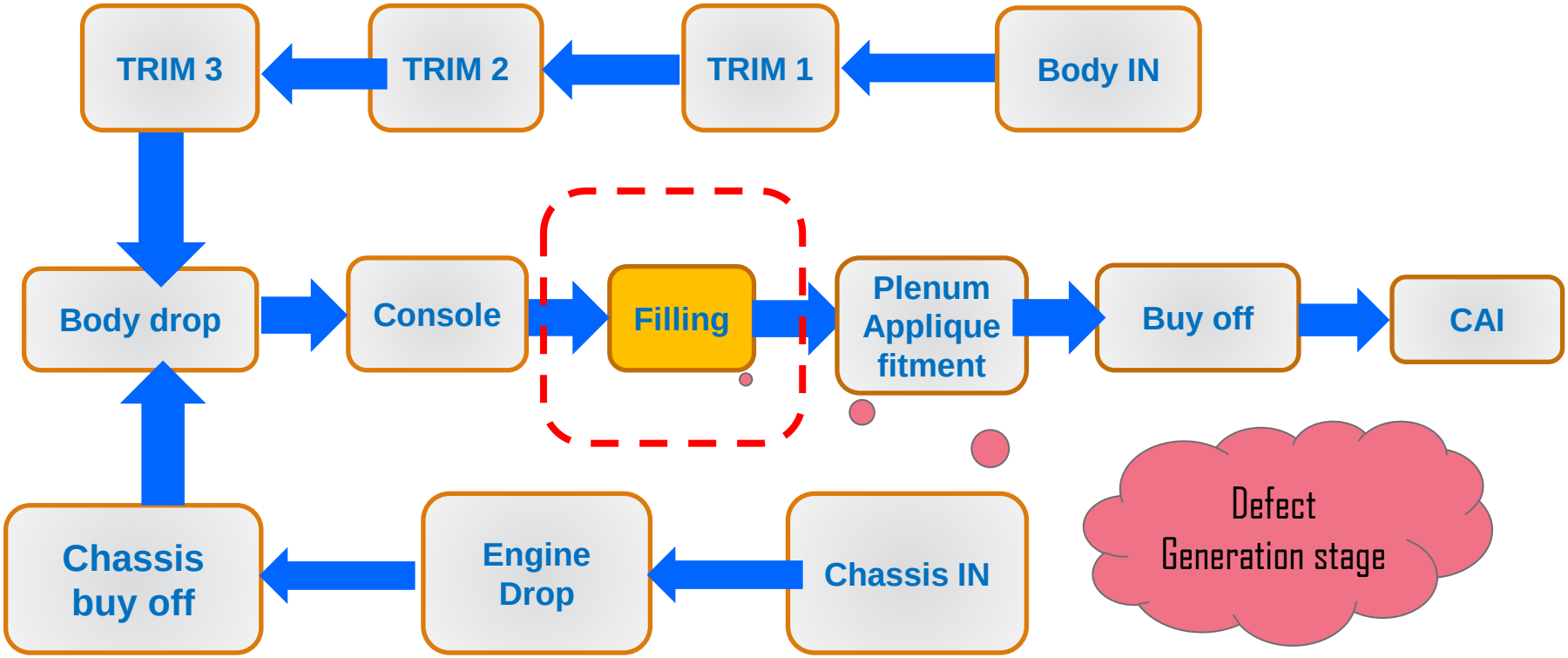
Total bolt not fitted concerns reported in last in last 3 months



Inference – Jack bolt slip concern reported for last 3 months, as per graph average 55/ month

Step 3. Understanding Current Situation

Flow Chart :- Observation of Process



Step 3. Understanding Current Situation

Observation of process

Collection of big bracket from storage bin

Collection of small bracket from storage bin

Collection of 4 M8 bolts from storage bin

Alignment of brackets on floor holes

Prefitment of all 4 Bracket bolts on floor holes

Tightening of bracket bolts using pneumatic until torque achieved

Collection of jack from storage rack

Tightening of jack by rotating lever

Step 3. Understanding Current Situation

Mahindra

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : SXTCF
Cell / BWT / Line: 11

Requested By- MQA | Date -05.11.2018 | Source -FAI | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Jack mounting bracket bolt not fitted

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- All scorpio | Model :Scorpio

Date of Incidence/ Assy :Aug, Sep, Oct | Shift / Time :all | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :
Vehicle rework done

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - | BWT . :

Does a control already exist ? ☐ YES ☐ NO

Stage / Post Analysis -

Man(Human Action)

Operator Name : All Jack stage associates
Is operator trained ? ☑ YES ☐ NO
Operator level (LL.U) L
OR Dexterity level(1...4) 3
Is operator under training ? ☐ YES ☑ NO
Is operator temporary for Stage ? ☐ YES ☑ NO
Bad rew ork ? ☐ YES ☑ NO

Method(process)

Main steps are respected ? ☐ YES ☑ NO
Is there a related key point in SOP/ SOS? ☑ YES ☐ NO
Are the reasons for key point know n ? ☐ YES ☑ NO
Are operating reqmnts respected ? ☐ YES ☑ NO
Is the defect part of the Buy off / Checkman list ? ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ? ☐ YES ☑ NO

Other

Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
Is the packaging standard defined for the part ? ☑ YES ☐ NO
Is the packaging standard respected ? ☑ YES ☐ NO
Is there any alert mechanism for failure ? ☐ YES ☑ NO

Machine

Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
Last calibration dates
Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☐ YES ☑ NO
Are machine parameters know n ? (Record in Analysis table) ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

Product Failure

In only one shift ? ☐ YES ☑ NO
In only one line ? ☑ YES ☐ NO
The problem follow s the vehicle ? ☑ YES ☐ NO
The problem follow s the part ? ☐ YES ☑ NO
Is it a repeat problem ? ☑ YES ☐ NO
Is the suspected parameter measurable ? (Record in Analysis table) ☑ YES ☐ NO

Material/ Consumable

Is the specified product used ? ☑ YES ☐ NO
Did it evolve / Change ? ☐ YES ☑ NO
Is the storage SOP respected ? ☑ YES ☐ NO
Is the shelf life / expiry date respected ? ☑ YES ☐ NO
Is the part as per FIFO? ☑ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
When analysis done It is found that,
1. Associate can not prefit bolts easily on floor panel
2. Carpet ovelapped observed on holes
3. Due to dark interior associate got dificulty while working
4. Due to cunjusted area associate can not holt tool properly fior tightening
5. Due this loose bracket jack can not be fitted.

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark

Is 4M Analysis Part 2 required ? ☐ YES ☑ NO

Is the substitute transport mode used ? ☐ YES ☑ NO | **Is the packaging defined ?** ☑ YES ☐ NO

Actions -

	Resp	Date	Status
1.Containment Action 1. DRF given to stage associates	Rahul	25.10.18	closed
2. Permanent Action - L1 analysis required	Resp	Date	Status

Analysed by Name/s -
Date - 25.10.18

Choice : ☐ WSS ☐ QC Story

4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3
Method	Operator aware of key point and check present point
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows Vehicle & Part
Material	Handling and shelf life of part adhered

NO Quick WIns

Step 3. Understanding Current Situation

Good & Not so good observations

- Issue observed only in Scorpio and not in Marazzo/Xylo.
- Associate is dexterity L3.
- Associate follows main step and key points.
- Torque used for tightening is as per control plan and torque sheet.
- Jack fitment stage associates working as per sos.
- Concern observed in both shifts at buy off.

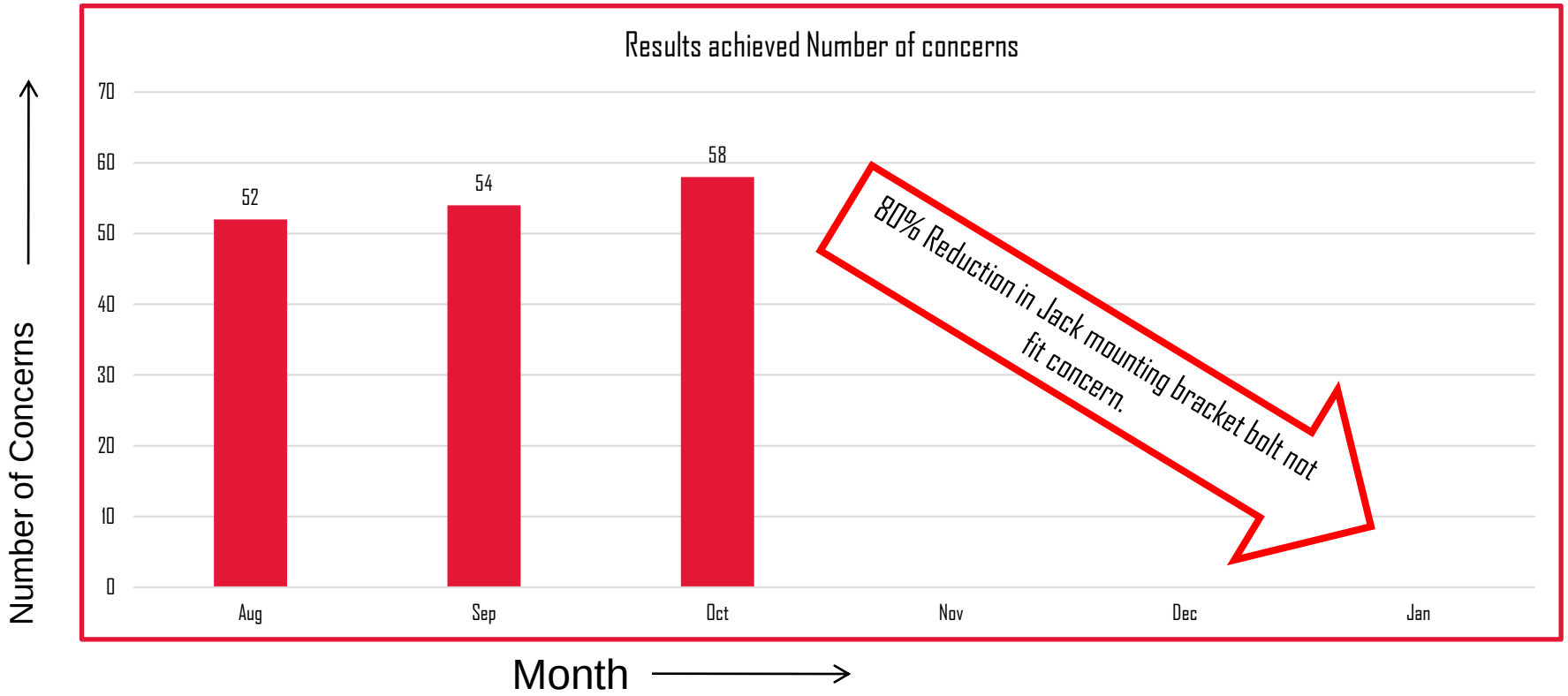
Step 3. Understanding Current Situation

Jack mtg location -

Jack mounting bracket
Part Number **0002AAG00040N**

4. Select Target

Present Level:	Average 55 concerns of Jack mounting bracket bolt not fit/ month.
Target:	80% reduction in Jack mounting bracket bolt not fit concern by Aug 18



5. Analysis

- **5.1 Identify Possible Causes : Through Brainstorming by CFT**

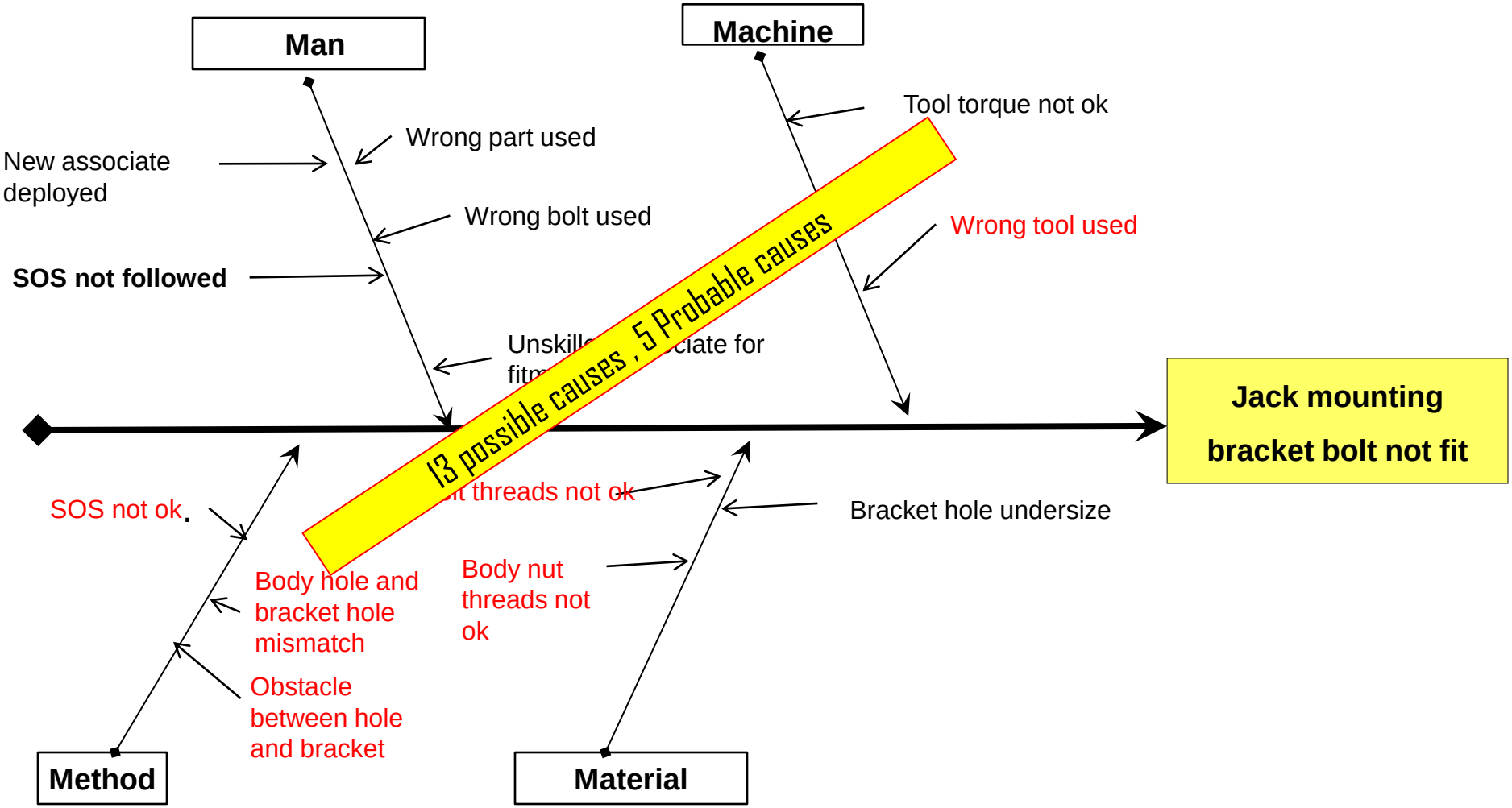
Sr. no	Possible Causes
1	New associate deployed at stage.
2	Bolt threads not ok
3	Body nut threads not ok
4	Wrong bolt used
5	Unskilled associate deployed
6	Wrong part used
7	Bracket hole undersize
8	SOS not followed
9	Obstacle between hole and bracket.
10	Wrong tool used
11	Tool torque not ok
12	SOS inadequate
13	Body hole and bracket hole missmatch

Brainstorming

Total 13 possible causes identified through brainstorming of CFT

5. Analysis

- **5.1 Identify possible causes: Fish Bone diagram**



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr. no	Possible Causes	Rahul	Amit	Santosh	Hemant	Gite	Nilesh	Ohol	Total
1	New associate deployed at stage.	0	1	0	0	3	3	0	**7**
2	Bolt threads not ok	3	3	9	1	9	0	9	**34**
3	Body nut threads not ok	9	3	9	0	9	0	9	**39**
4	Wrong bolt used	0	0	0	0	0	0	0	**0**
5	Unskilled associate deployed	1	1	3	0	0	1	3	**9**
6	Wrong part used	3	0	0	3	0	3	0	**9**
7	Bracket hole undersize	3	3	0	3	9	3	3	**24**
8	SOS not followed	3	1	3	3	1	3	1	**15**
9	Obstacle between hole and bracket.	9	3	3	3	9	3	9	**39**
10	Wrong tool used	3	1	1	3	1	3	1	**13**
11	Tool torque not ok	3	1	3	3	1	3	1	**15**
12	SOS inadequate	9	9	3	0	0	3	9	**33**
13	Body hole and bracket hole missmatch	9	9	1	3	9	1	9	**41**

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 33

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. no	Possible Causes	Reasons for not considered
1	New associate deployed at stage.	All time trained associate is deployed on stage
4	Wrong bolt used	Bolt storage location checked and part nuber checked found ok
5	Unskilled associate deployed	All time trained associate is deployed on stage
6	Wrong part used	Correct part availble on stage
7	Bracket hole undersize	50 parts checked found ok
8	SOS not followed	Trained associates deployed on stage
10	Wrong tool used	Checked on stage, other tools can not be used
11	Tool torque not ok	Torque found ok

5. Analysis

5.1 Identify Probable causes : High severity causes selected during brain storming for test of hypothesis

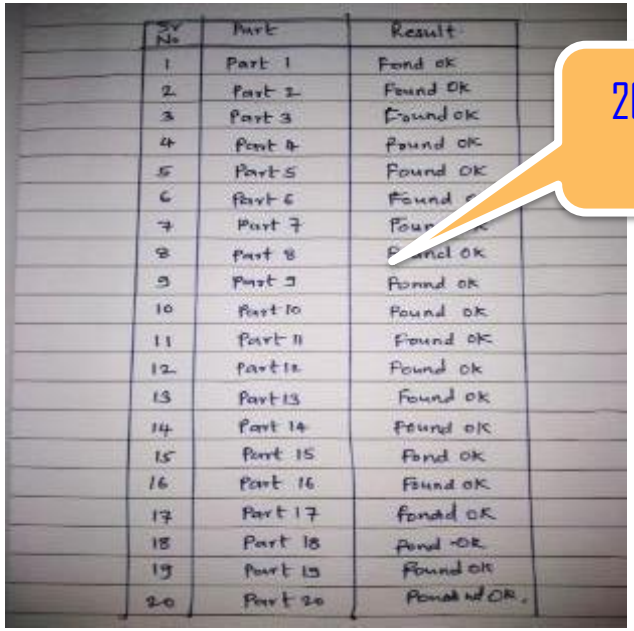
Sr. no	Probable causes
1	Bolt threads not ok
2	Body nut threads not ok
3	Obstacle between hole and bracket.
4	SOS inadequate
5	Body hole and bracket hole mismatch

From 13 Possible causes 5
are selected as probable causes

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Bolt threads not ok	Checking bolt threads using M8 nut By checking storage	20 parts from different bins checked with M8 nut which is found ok Storage checked found ok as per std practice	Hypothesis invalid



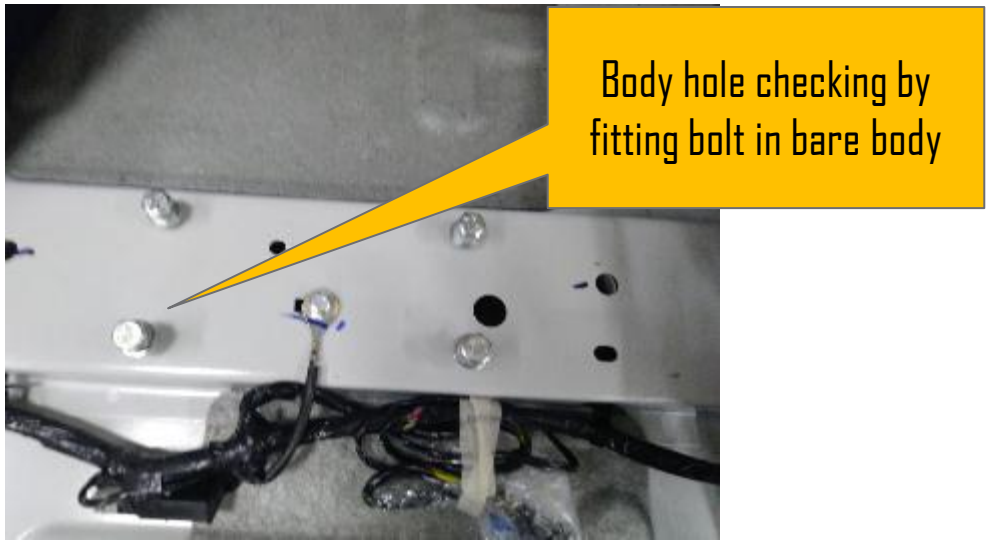
Sr No	Part	Result
1	Part 1	Found ok
2	Part 2	Found OK
3	Part 3	Found ok
4	Part 4	Found OK
5	Part 5	Found OK
6	Part 6	Found
7	Part 7	Foun
8	Part 8	Found ok
9	Part 9	Found ok
10	Part 10	Found ok
11	Part 11	Found OK
12	Part 12	Found ok
13	Part 13	Found ok
14	Part 14	Found ok
15	Part 15	Found ok
16	Part 16	Found OK
17	Part 17	Found OK
18	Part 18	Found OK
19	Part 19	Found OK
20	Part 20	Found ok OK.

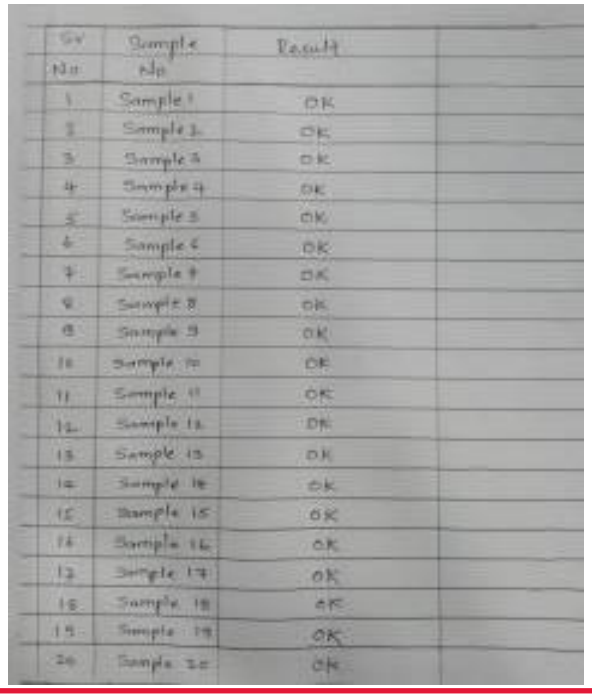
Inference:- As out of 20 parts all parts Found ok hence Probable cause is invalid.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Body nut threads not ok	Inspection of body nut by inserting bolt	20 vehicles checked for body nut threads not ok	**Hypothesis Invalid**





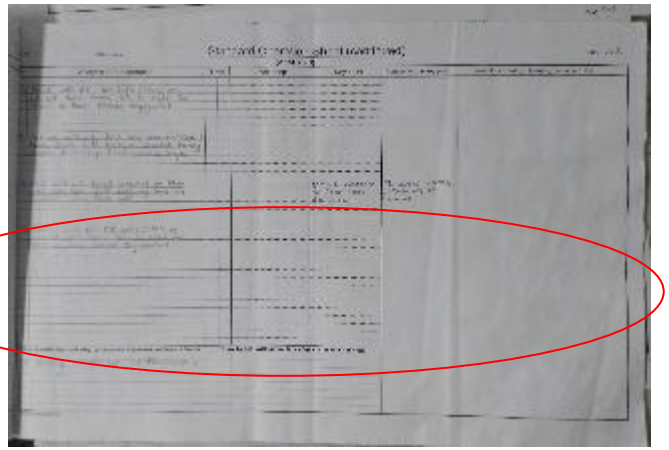
Sr No	Sample No	Result
1	Sample 1	OK
2	Sample 2	OK
3	Sample 3	OK
4	Sample 4	OK
5	Sample 5	OK
6	Sample 6	OK
7	Sample 7	OK
8	Sample 8	OK
9	Sample 9	OK
10	Sample 10	OK
11	Sample 11	OK
12	Sample 12	OK
13	Sample 13	OK
14	Sample 14	OK
15	Sample 15	OK
16	Sample 16	OK
17	Sample 17	OK
18	Sample 18	OK
19	Sample 19	OK
20	Sample 20	OK

Inference:- As out of 20 parts all parts Found ok hence Probable cause is invalid.

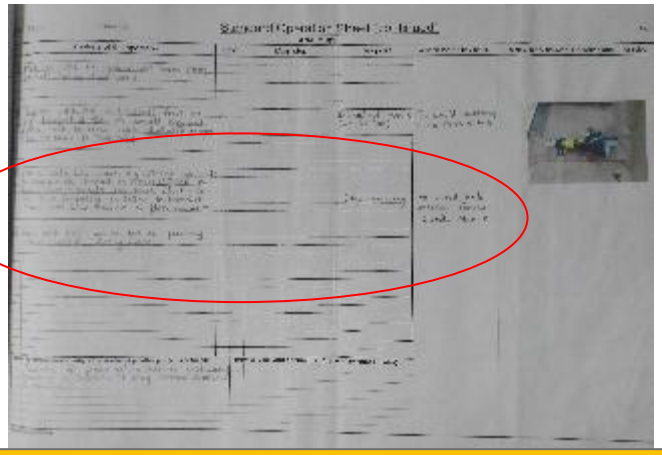
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	SOS inadequate	Going brief through Jack mounting bracket fitment SOS	Current SOS says Install Jack mounting brackets on floor by prefitting bolts Installing Jack in bracket Tightening of jack mounting bolts(4nos) As per this SOS bracket bolts tightening under seat is very difficult and causes bolt not fit issues	**Hypothesis Valid**



Prefitment of Jack mounting bolts

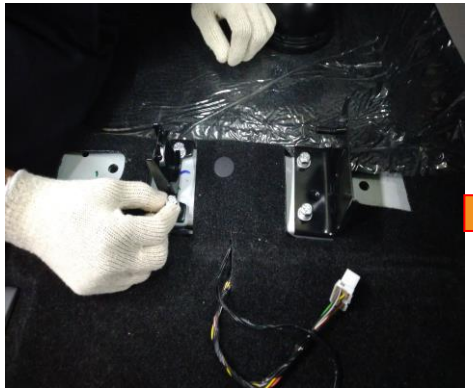


Tightening of Jack mounting bracket bolts after Jack tightening in brackets

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	SOS inadequate	Going brief through Jack mounting bracket fitment SOS	Current SOS says Install Jack mounting brackets on floor by prefitting bolts Installing Jack in bracket Tightening of jack mounting bolts(4nos) As per this SOS bracket bolts tightening under seat is very difficult and causes bolt not fit issues	**Hypothesis Valid**



Prefitment of Jack mounting bolts

Fitment of jack in jack mounting bracket before bracket bolts tightening

Tightening of Jack mounting bracket bolts after Jack tightening in brackets

Due to poor access to bolt associate either skip this bolt or bolt gets slipped or half torque.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	Obstacle between hole and bracket.	Checking of hole whether it is getting covered by other part which is causing obstacles for bolt prefitment and tightening	1. 30 vehicles observed then in 11 vehicles found that Floor carpet is covering bracket mounting hole which results into bolt not fit. Carpet is getting stretched while LH front scuff plate fitment	**Hypothesis Valid**

Scuff plate fitment process

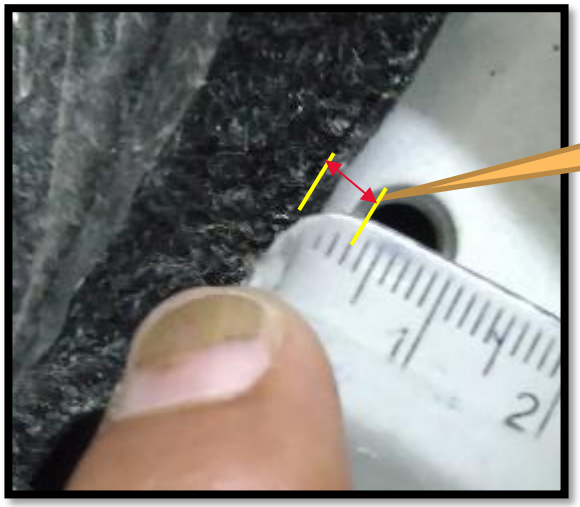
Scuff plate fitment process

- Jack mounting bracket carpet cutout doesn't remain in its position .
- As shown in picture it gets stretched on bracket mounting bolt hole.
- At the time of scuff plate fitment this cut out gets pulled away from scuffplate

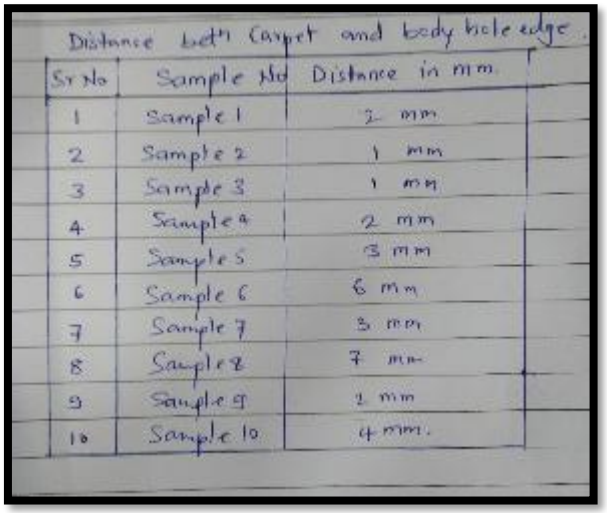
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	Obstacle between hole and bracket.	Checking carpet cut out edge to hole edge distance	10 vehicles checked and 8 vehicles found Carpet cut out for jack mounting bracket is not in place. Carpet cut out and Body hole Edge to edge distance is very less found below 5mm in every vehicle before	**Hypothesis Valid**



Carpet cut out and hole distance



Distance betn Carpet and body hole edge

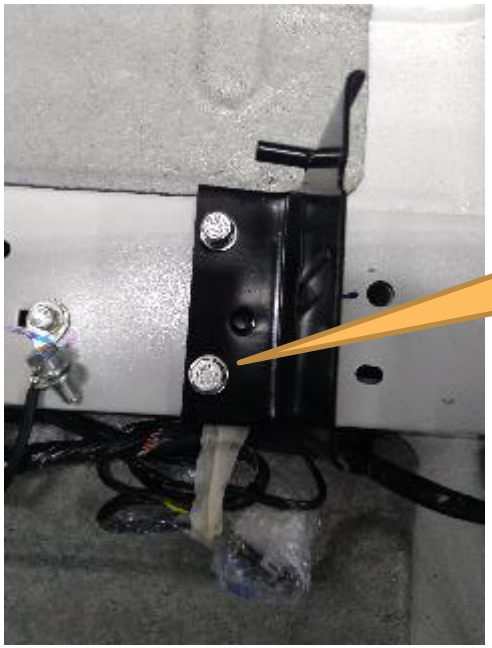
Sr No	Sample No	Distance in mm
1	Sample 1	2 mm
2	Sample 2	1 mm
3	Sample 3	1 mm
4	Sample 4	2 mm
5	Sample 5	3 mm
6	Sample 6	6 mm
7	Sample 7	3 mm
8	Sample 8	7 mm
9	Sample 9	1 mm
10	Sample 10	4 mm.

Checking Carpet cutout and hole distance

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	Body hole and bracket hole mismatch	Prefitment of one bracket bolts on b 10 different bodies and 10 brackets in one body by aligning bracket hole and body hole	Testing done on 10 bare bodies and no body hole and bracket hole found mismatch.	**Hypothesis invalid**



Jack Bracket hole Body hole alignment

Sr No	Sample No	Result
1	Sample1	ok
2	Sample 2	ok
3	Sample 3	ok
4	Sample 4	ok
5	Sample 5	ok
6	Sample 6	ok
7	Sample 7	ok
8	Sample 8	ok
9	Sample 9	ok
10	Sample 10	ok.

Inference:- As out of 10 bodies 10 bodies Found ok hence Probable cause invalid.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1	Bolt threads not ok	20 parts from different bins checked with M8 nut which is found ok Storage checked found ok as per std practice	Hypothesis Invalid
2	Body nut threads not ok	20 vehicles checked for body nut threads not ok	Hypothesis Invalid
3	Obstacle between hole and bracket.	1. 30 vehicles observed then in 11 vehicles found that Floor carpet is covering bracket mounting hole which results into bolt not fit. Carpet is getting stretched while LH front scuff plate fitment 2. 10 vehicles checked and 8 vehicles found Carpet cut out for jack mounting bracket is not in place. Carpet cut out and Body hole Edge to edge distance is very less found below 5mm every vehicle before	Hypothesis Valid
4	SOS inadequate	As per Current SOS Install Jack mounting brackets on floor by prefitting bolts Installing Jack in bracket Tightening of jack mounting bolts(4nos) As per this SOS bracket bolts tightening under seat is very difficult and causes bolt not fit issues	Hypothesis Valid
5	Body and bracket hole mismatch	Testing done on 10 bare bodies and no body hole and bracket hole found mismatch.	Hypothesis Invalid

WHY - WHY ANALYSIS

5.3 Identify the root causes:

Sr No	Valid Probable Cause	WHY	WHY	WHY	WHY	Required Actions
1	Obstacle between hole and bracket.	Jack mounting brackets bolts slip or half torque while delivering torque	Prefitment of jack mounting bracket bolts not ok	Carpet edge from front side found in between hole and bolt	Carpet position not ok	Shift the carpet edge toward front
2	SOS inadequate	part fitment sequence is wrong in SOS	Associates training as per SOS	Difficult to tighten Jack bracket bolts under seat as limited access	Tightening sequence inadequate	Change the sequence

6. Implement the corrective measures

Possible Remedies Identified

Sr. No.	Probable Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Carpet position not ok	Shift the carpet edge toward front	Low	Low	Yes	Low	**Accepted**
2	Tightening sequence inadequate	Change the sequence	Low	Low	Yes	medium	**Accepted**

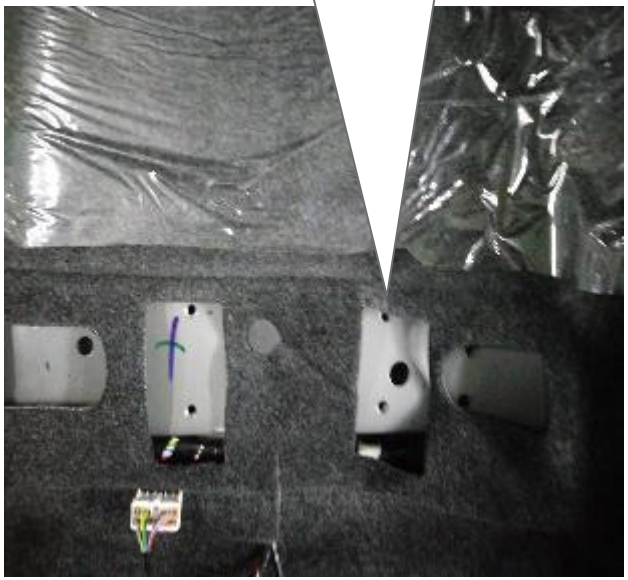
6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action Plan	Target Dt	Resp.	Status
2	Obstacle between hole and bracket.	Shift the carpet edge toward front by enlarging cut out by 10 mm	14/09/2018	Swapnil Patil	Closed

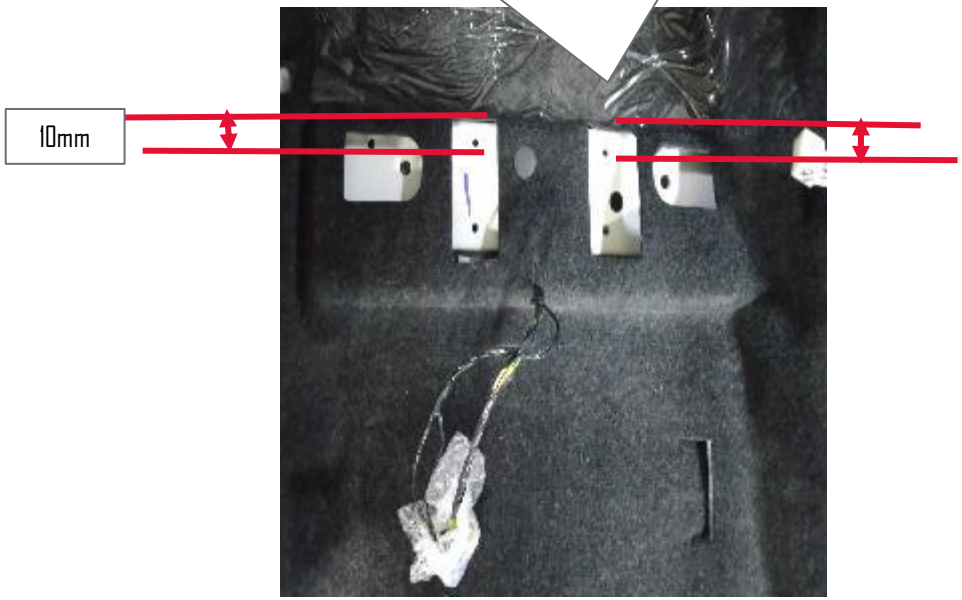
BEFORE

Carpet edge Shifted on jack mounting bracket hole



AFTER

Carpet cut out increased by 10 mm to shift ed from bracket hole



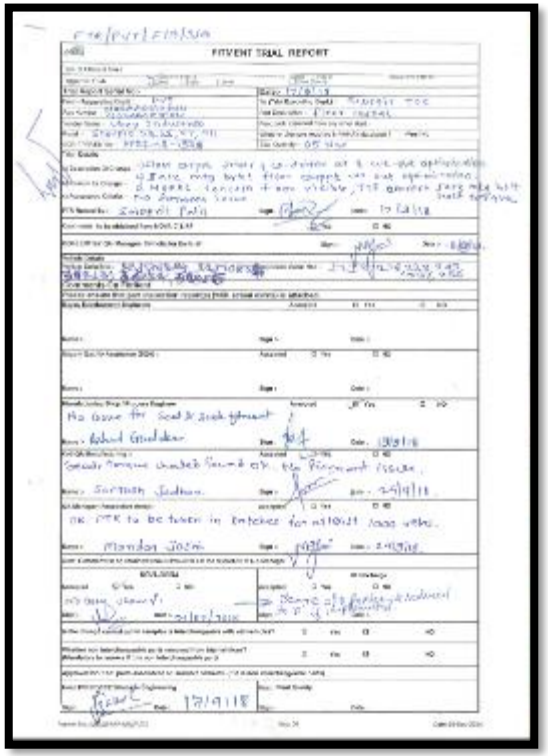
6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action Plan	Target Dt	Resp.	Status
2	Obstacle between hole and bracket.	Shift the carpet edge toward front by enlarging cut out by 10 mm	17/09/2018	Swapnil Patil	Closed

Carpet cut out increased by 10 mm at front side of vehicle

FTR



6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action Plan	Target Dt	Resp.	Status
2	Obstacle between hole and bracket.	Shift the carpet edge toward front by enlarging cut out by 10 mm	14/11/2018	Swapnil Patil	Closed

Carpet cut out increased by 10 mm at front side of vehicle

EN CHANGE:- W110-EN-A003-000117

Mahindra

Engineering Notice Release: W110-EN-A003-000117
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 29-Oct-2018
Initiated Date 13-Nov-2018

Engineering Notice No : W110-EN-A003-000117
Released Date : 14-Nov-2018 10:08
Reference Change Req No : DCR041829
Location : AD-NASIK
Implementation Location : M&M-Nasik
Engineering Notice Description : OPTIMISATION OF FLOOR CARPET CUTOUT FOR DRIVER & CODRIVER SEAT MOUNTING AREA
Reason : TO ADDRESS INTERNAL CONCERN OF GNOVAC OF FLOOR BODY VISIBLE
Quantity : -(Regular)
Reason Code : Quality Improvement
Priority : Urgent
Note : TO BE IMPLEMENTED FOR PRODUCTION
Affected Models : SCORPIO S3 S5 S7 S11,EXPORT W PASSANGER ONLY,
Is EN Re Released? :
Parent EN NO :
Life Cycle State : Released
Owner Name : Joshi Girish N (310471)

Engineer : Joshi Girish N (310471)
Approved By : 211714-PATHAN YOGESH
Authorized By : HEAD-PVT-AD_202475-RAUT AVEDH

Legends

Add New : A Part/Assy that is getting released through EON for the very first time

Modified : A Part/Assy that is already released through an EON and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

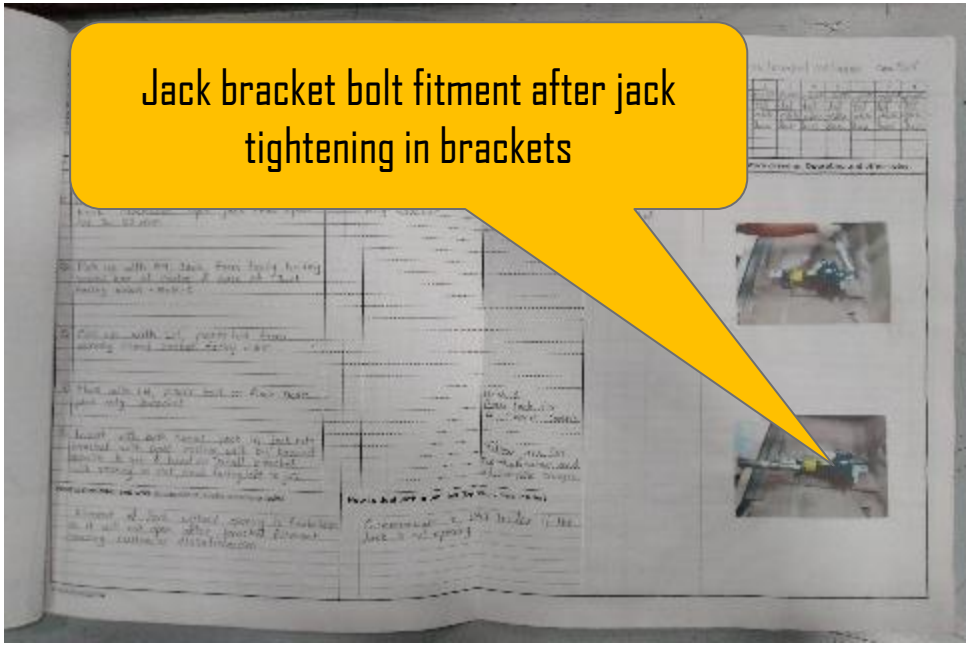
ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

6. Implement the corrective measures

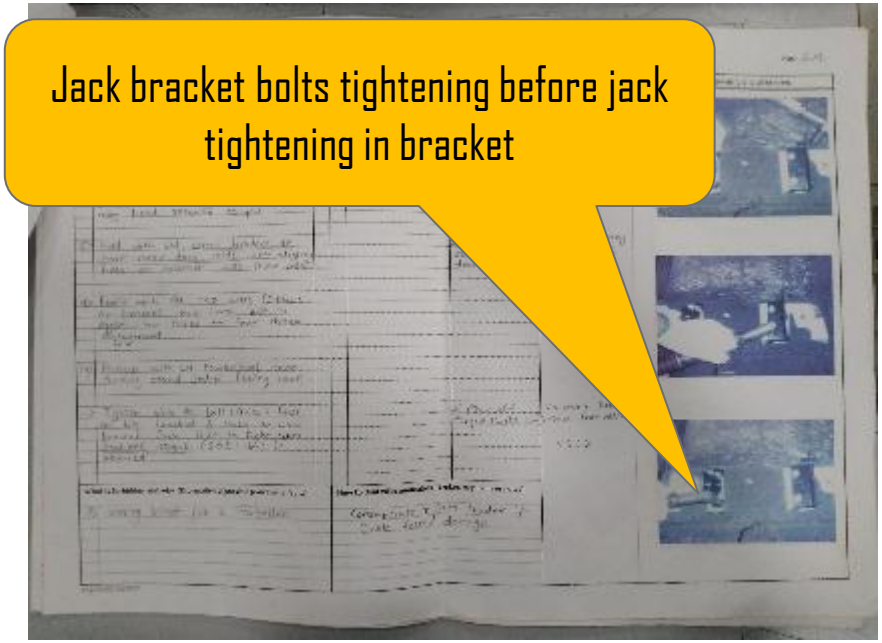
6.1 Actions Decided

No	Root Cause	Action Plan	Target Dt	Resp.	Status
2	SOS inadequate	Correct the SOS	15/01/2019	Rahul Gadekar	Closed

BEFORE



AFTER



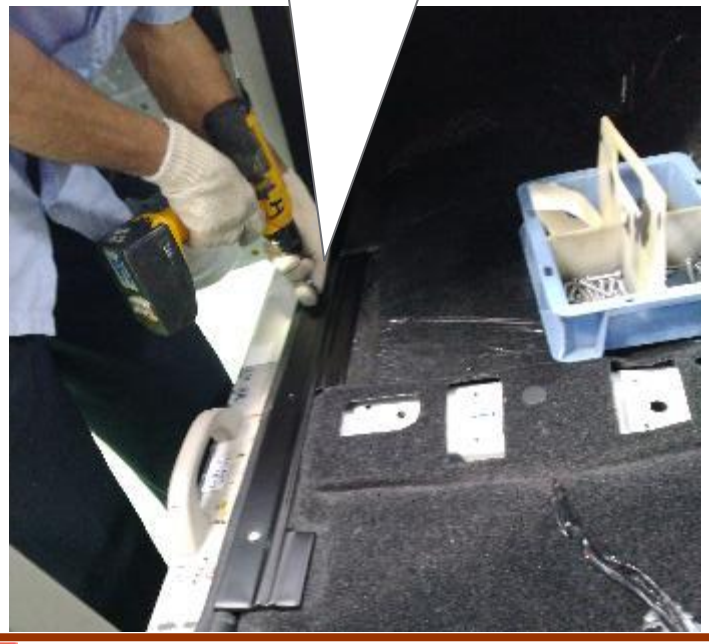
6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action Plan	Target Dt	Resp.	Status
2	Obstacle between hole and bracket.	Shift the carpet edge toward front	14/04/2019	Tarit Nandi	Closed

BEFORE

Fitment of LH front scuff plate without adjusting carpet



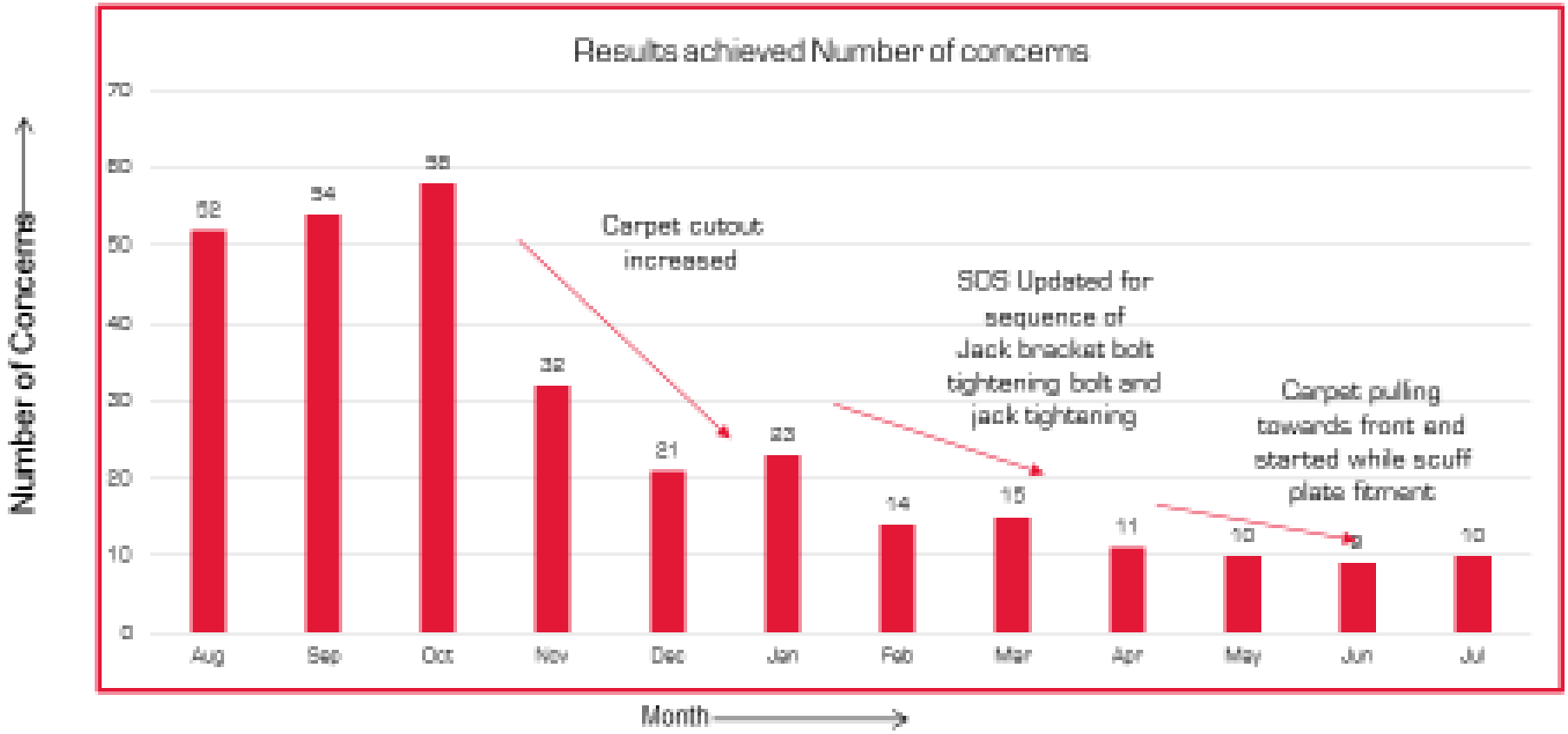
AFTER

Fitment of LH front scuff plate by pulling and pressing carpet towards front side of vehicle

7. Check & confirm the results & effects

80% reduction Jack bolt not fit concern

7.1 Verify Results:



Inference – Jack mtg bkt not fit after taking PCA actions , 81% reduction in concern observed

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	Shift the carpet edge toward front by enlarging cut out by 10 mm	14/09/2018	Swapnil Patil	Closed
2	Correct the SOS	15/01/2019	Rahul Gadekar	Closed
3	Fitment of LH front scuff plate by pulling and pressing carpet towards front side of vehicle	14/04/2019	Rahul Gadekar	Closed
4	PFMEA updation (Rev No –28)	21.11.18	Suhas Aher	CLOSED
5	Process audit check sheet updation (Rev No -6)	10.03.19	Rahul Gadekar	CLOSED
6	OJT Training to stage operator	11.12.18	Rahul Gadekar/ Hemant Khare	CLOSED

7.2 Tangible & Intangible benefits:

Tangible benefits:

Tangible Benefits -

- More than 80% reduction achieved for jack mtg bolt not fitted concern.
- Rework cost is reduced by approx. Rs. 21123.61/Annum.

Cost approval from Industrial engineer.

DHANOKAR RAJESH

Re:Pls approve following cost estimation

To GADEKAR RAHUL

okay approved

Sent from my Redmi Note 4

On GADEKAR RAHUL <GADEKAR.RAHUL2@mahindra.com>, 01-Dec-2019 2:27 PM wrote:

Cost estimation for Jack mth bkt bolt change		
Description	Units	
Rework time (Approx)	Min	90
Avg Concerns/Day	Nos	1
Total Required Rework time per day	Min	90
Effective working time	Min	465
Operator required /day	nos	0.193548387
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	62.90322581
Avg no. of working days	days	25
Operator cost /month	Rs	1572.580645
Operator Cost/year	Rs	18870.96774
Bolt cost per veh	Bolts / veh.	1
Bolt cost	Rs/bolt	3.61
Daily bolts rejection	No bolts /day	2
Cost impact per bolt on change	Rs/day	7.22
Project annual volume	Rs / Year	2252.64
Total Rework cost	Rs / Year	21123.61

Rahul

7.2 Tangible & Intangible benefits:

Intangible benefits:

1. **Repetitive work eliminated hence improved Moral**
2. **Hesitation while work is reduced.**
3. **Line flow become smooth due to no rework**

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S N	Document	Target date	Responsibility	Status
1	PFMEA updation (Rev No –28)	21.11.18	Suhas aher	closed
2	Process audit check sheet updation (Rev No -9)	15.04.19	Rahul Gadekar	closed
3	Jack fitment SOS Updation (Rev No - 10)	15.01.19	Rahul Gadekar	closed
4	Scuff plate fitment SOS updation (Rev No 12)	14.04.19	Ganesh Bendale	Closed

8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	On job Training given to Scuff plate fitment associate as per new method.	05.03.19	Rahul Gadekar	Closed

8. Standardization

8.1 Updation of documents :-

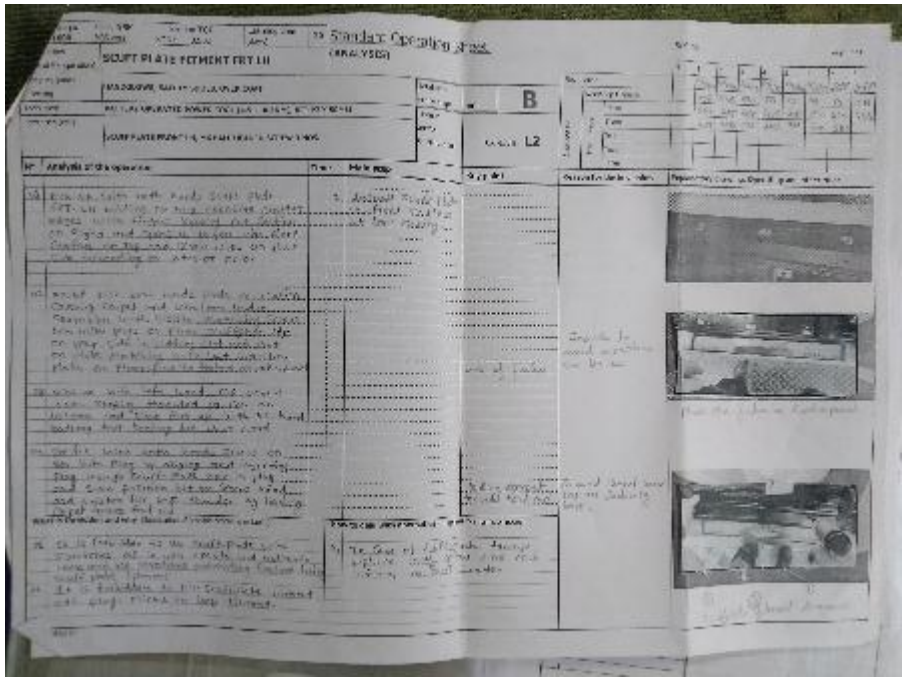
PFMEA Updation

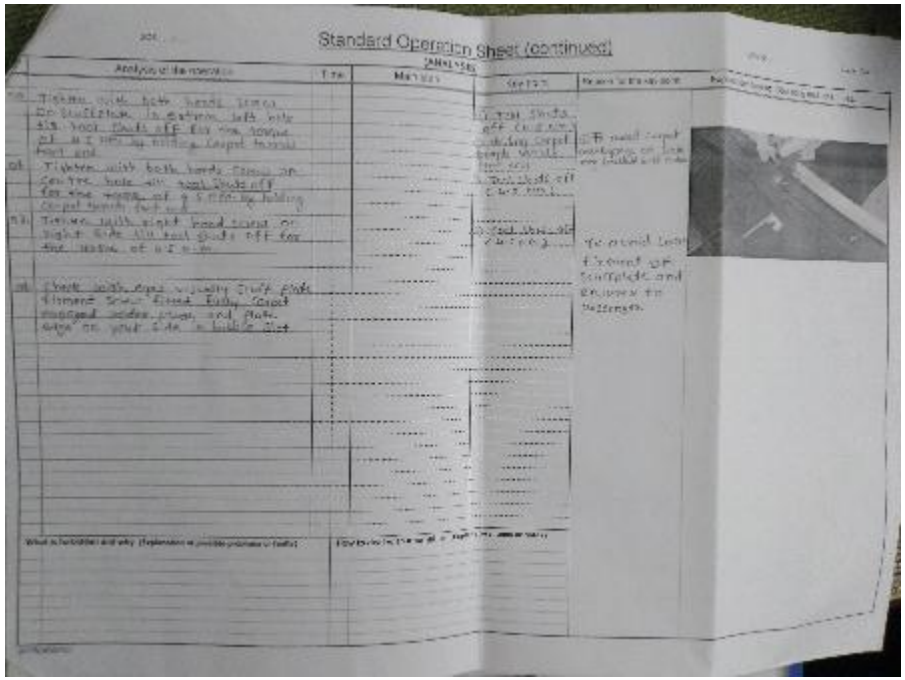
Plant : Nashik			Model :Scorpio														Cross Functional Team			
Part Nos.:			Part Name/ Process:														Sohas Aher	Bhab Ashale		
Latest Change Letter:			Optn No.: 4960														Vikrant Ghule	Santosh Jadhav		
PUF Shop :Scorpio TCF			Optn Despin:Jack mtg bracket fitment														Sagali More			
Module/ Cell :Final			MIC Name & No.:																	
Process Function	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	Sev	Class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
Jack mtg bracket fitment	Fitment of jack mtg. Bracket Bolts	1) Bolt half torque. (Bug Off concern)	Mfg. & Assg: rattle noise & rework..(5) End user: Rattle Noise..(4)	4		Operator cross fit the bolt	1. Operator trained to work as per SOP 2. Forbidden point added in SOS	8	Check point given at bug off.	7	168	Jack bolt cut out to be increased by 10 mm towards front end	Swapnil Patil 14.11.2018	Action completed				
		2) Bolts under torqued.	Mfg. & Assg: rattle noise & rework..(5) End user: Rattle Noise..(4)	5		Improper power tool calibration	Periodic Tool Calibration	2	Torque audit.	7	70							
						Use of wrong power tool	Dedicated power tool.		Identification sticker on Power tool									
		3) Bolts overtorqued.	Mfg. & Assg: Bolt may get sliped, thread damaged and rework.(5) End user: Rattle Noise..(4)	4		Improper power tool calibration	Periodic Tool Calibration	3	Torque audit.	7	84							
						Use of wrong power tool	Dedicated power tool.		Identification sticker on Power tool									

8. Standardization

8.1 Updation of documents:

SOS updated as per Process change

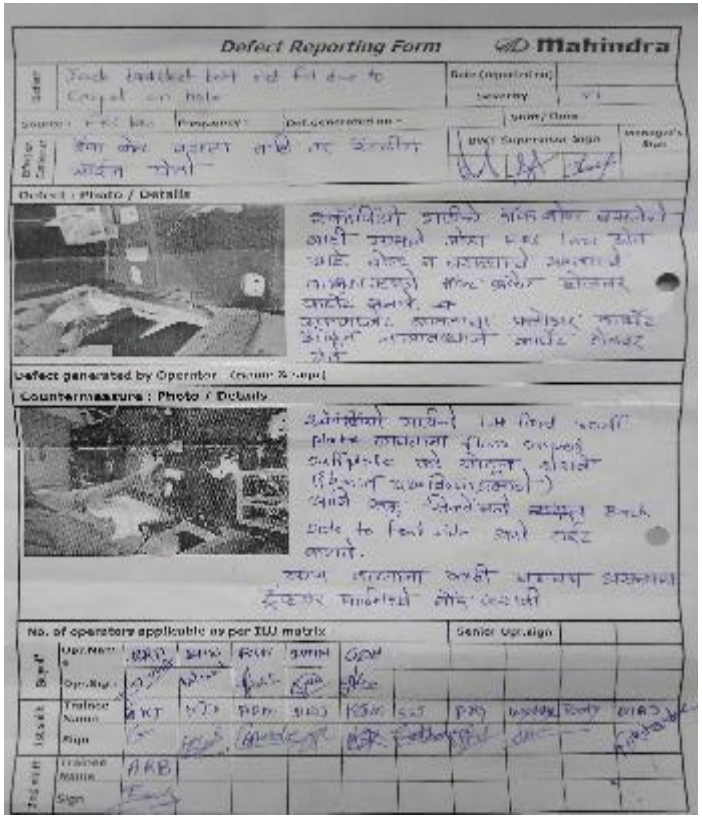




8. Standardization

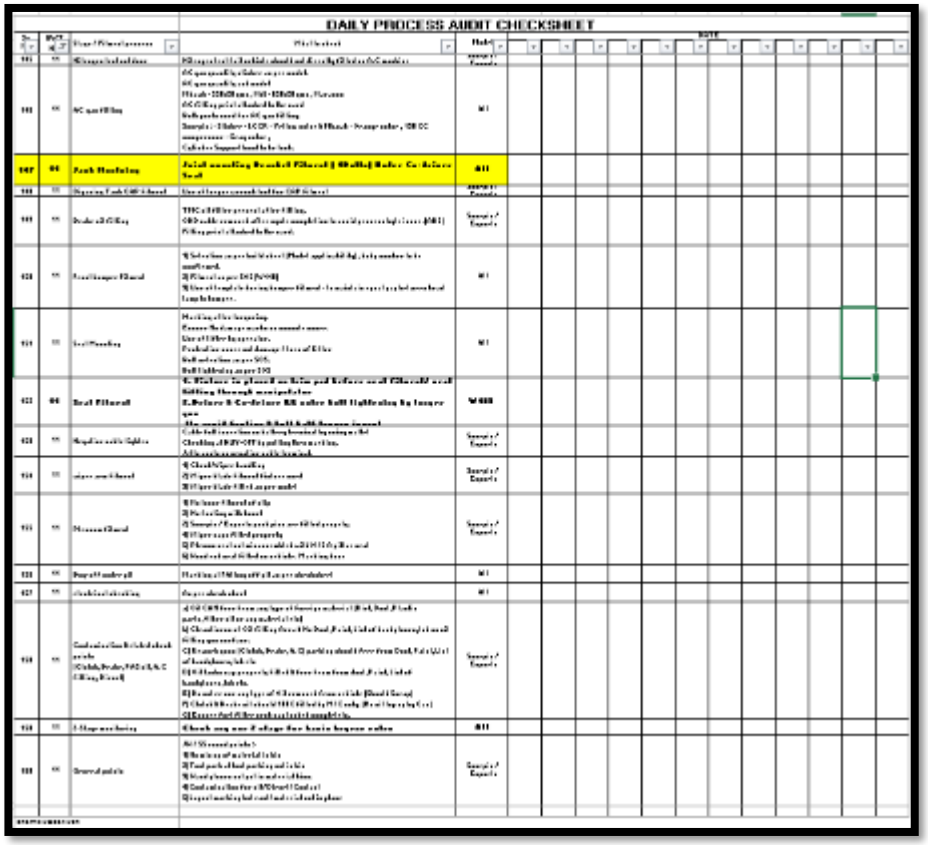
8.2 Training:

OPL given to stage associates for working as per New process & awareness about actions taken

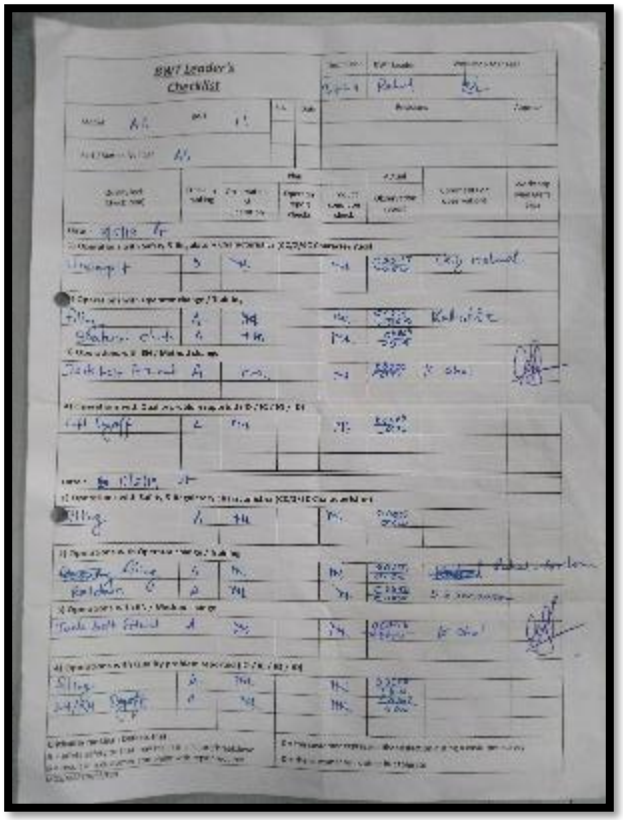


8. Standardization

8.3 Additional Audit check



Process audit checksheet



Superwiser checksheet

9. Summarize & Plan future actions

9.1 Conclusion:

By Doing changes in Carpet assy and by changing scuff plate tightening method as well as jack fitment process we are able to reduce Jack bolt not fit concern by 80%

9.2 Horizontal Deployment:

Nil

9.3 Lessons Learnt:

Practical, GAMBA activity and Deep process study is useful for solving concerns

Things Gone Wrong :

During Project Phase Effect of jack bolt sequence was not considered properly.

Things Gone Right:

By detailed analysis of Process parameters and carpet specification we able to reduce concern by 80%.

Thank You!